

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	35.3	16.8	2.4	0.6	0.2	44.7
U3 - Toxicity prior Beta(0.15, 0.85)	35.4	15.6	2	0	0.1	46.9
U2 - Toxicity prior Beta(0.3, 0.7)	35.7	16.4	3.2	0.6	0.4	43.7
U3 - Toxicity prior Beta(0.3, 0.7)	35.8	16.6	1.8	0.2	0.2	45.4
U2 - Toxicity prior Beta(0.5, 0.5)	36.2	19.2	4.2	0.9	0.5	39
U3 - Toxicity prior Beta(0.5, 0.5)	37.9	17.6	3.4	0.4	0.2	40.5
U2 - Toxicity prior Beta(1, 1)	36.7	19	4.9	0.7	0.5	38.2
U3 - Toxicity prior Beta(1, 1)	39.3	18.2	3	0.4	0.2	38.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.6	6.3	2.5	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	15.1	5.9	2.3	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	14.5	6.4	2.6	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	15	6	2.3	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	13.7	6.9	3.2	1.8	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	14.4	6.4	3.1	1.7	0.6	
U2 - Toxicity prior Beta(1, 1)	13.6	7	3.4	1.8	0.7	
U3 - Toxicity prior Beta(1, 1)	14.3	6.5	3.1	1.7	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.4	0.8	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.3	0.7	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.5	0.8	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.3	0.7	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3.4	0.9	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.3	0.9	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	1.2	3.4	0.9	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	1.2	3.3	0.9	0.7	0.2	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.2	18.3	39.3	13.9	2.3	1
U3 - Toxicity prior Beta(0.15, 0.85)	30.8	19.1	35.4	11.9	1.8	1
U2 - Toxicity prior Beta(0.3, 0.7)	23.1	17.1	41.2	15.1	2.7	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	29.9	18	36.1	13.1	2.1	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	20.7	14	44.4	17.1	3	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	25.6	16.6	40.3	14.5	2.2	0.8
U2 - Toxicity prior Beta(1, 1)	20.4	14.8	43.5	17.5	3.2	0.6
U3 - Toxicity prior Beta(1, 1)	25.2	15.9	40.8	15.2	2.3	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	7	9	3.8	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	7.2	8.7	3.5	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	8.9	6.9	9.1	3.9	1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	7.1	8.7	3.7	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	6.8	9.8	4.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	7	9.6	4	0.9	
U2 - Toxicity prior Beta(1, 1)	7.6	6.9	9.9	4.4	1.2	
U3 - Toxicity prior Beta(1, 1)	8.1	7.1	9.7	4	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.2	1.7	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.6	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.7	0.9	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.2	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.8	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3	1.7	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	2.9	1.8	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.5	2.9	1.8	0.9	0.2	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	46.7	26.9	12.3	2.1	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	12.4	53.6	21.1	8.7	1.7	2.5
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	46.4	26.9	13.1	2.6	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	11.5	52.4	22.2	9.8	2	2.1
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	45.3	26.3	14.3	3.4	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	11	51.5	22	11.5	2.1	1.9
U2 - Toxicity prior Beta(1, 1)	8.4	45.3	26.3	14.6	3.6	1.8
U3 - Toxicity prior Beta(1, 1)	10.2	52.9	22.1	10.9	2.1	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.6	11.3	6.8	3.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	6.8	12.2	6.4	3.4	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	6.5	11	6.9	4.1	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	6.6	11.9	6.5	3.6	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	5.6	11.2	7.1	4.4	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	5.8	12.1	6.8	3.9	1.2	
U2 - Toxicity prior Beta(1, 1)	5.5	11.2	7.2	4.4	1.5	
U3 - Toxicity prior Beta(1, 1)	5.6	12.1	6.9	3.9	1.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	1.8	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.8	1.6	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	1.8	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.8	1.7	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.3	1.8	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.6	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.3	3.3	1.8	1.2	0.4	
U3 - Toxicity prior Beta(1, 1)	0.4	3.4	1.7	1.1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.7	6.6	17.5	52	17	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	7.8	16.9	52.1	13.4	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	5.2	5.6	15.3	55	18.7	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	7.7	15	52.6	15.3	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	5.2	5.1	13.5	55.2	20.8	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	7.9	6.5	14.3	54.1	16.9	0.3
U2 - Toxicity prior Beta(1, 1)	4.7	5	14.3	54.9	20.8	0.3
U3 - Toxicity prior Beta(1, 1)	8	6.9	13.5	54.6	16.8	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.2	4.6	6.7	10.6	3.9	
U3 - Toxicity prior Beta(0.15, 0.85)	4.5	4.6	6.8	10.7	3.4	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	4.4	6.6	10.8	4.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.4	4.4	6.5	10.9	3.7	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	4.1	6.3	11.4	4.5	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	4.2	6.4	11.5	4	
U2 - Toxicity prior Beta(1, 1)	3.7	4	6.2	11.4	4.6	
U3 - Toxicity prior Beta(1, 1)	3.9	4.2	6.3	11.6	4.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.5	2.4	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.5	2.4	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.5	2.4	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.5	2.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.3	0.7	
U2 - Toxicity prior Beta(1, 1)	0.1	2.7	1.4	2.3	0.9	
U3 - Toxicity prior Beta(1, 1)	0.2	2.8	1.5	2.3	0.7	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	38.4	25.4	19.7	10.7	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	8.1	43.6	24	17.6	6.4	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	5.2	37.4	25	20.8	11.3	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	44.2	23.1	17.9	6.4	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	5.6	36.7	24.8	20.5	12.1	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	8	44.1	22.2	18.2	7.1	0.4
U2 - Toxicity prior Beta(1, 1)	6	35.5	24.9	20.9	12.3	0.4
U3 - Toxicity prior Beta(1, 1)	8.8	43.5	22	18.3	7.1	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.9	7.3	7	7.1	4.7	
U3 - Toxicity prior Beta(0.15, 0.85)	4	7.9	7.3	6.9	3.9	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	7.1	7	7.2	4.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4	7.8	7.1	7	4.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.5	6.9	6.9	7.5	5.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	7.7	7.1	7.1	4.4	
U2 - Toxicity prior Beta(1, 1)	3.5	6.7	7	7.6	5.2	
U3 - Toxicity prior Beta(1, 1)	3.6	7.7	7.1	7.1	4.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	3.1	1.9	2.2	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	2	2.2	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	3.1	1.9	2.3	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.9	2.1	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.9	2.3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3	1.9	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.9	2.4	1.2	
U3 - Toxicity prior Beta(1, 1)	0.2	3	1.9	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	63.5	17.5	4.9	1.9	0.5	11.7
U3 - Toxicity prior Beta(0.15, 0.85)	66.7	16.2	3.7	1.2	0.5	11.7
U2 - Toxicity prior Beta(0.3, 0.7)	62.7	19.4	4.6	1.9	0.5	10.9
U3 - Toxicity prior Beta(0.3, 0.7)	64.2	18.8	3.8	1.4	0.4	11.4
U2 - Toxicity prior Beta(0.5, 0.5)	60.4	20	6.6	2.6	0.6	9.8
U3 - Toxicity prior Beta(0.5, 0.5)	64.1	20.5	4.3	1.4	0.4	9.3
U2 - Toxicity prior Beta(1, 1)	59.3	21.6	6.4	2.7	0.7	9.3
U3 - Toxicity prior Beta(1, 1)	63.6	20.3	4.7	1.4	0.4	9.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.5	6.1	2.7	1.3	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	18.1	6	2.4	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	17.2	6.3	2.8	1.4	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	17.8	6.2	2.5	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	16	6.7	3.4	1.6	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	16.6	6.8	3.1	1.4	0.4	
U2 - Toxicity prior Beta(1, 1)	15.7	6.9	3.5	1.7	0.5	
U3 - Toxicity prior Beta(1, 1)	16.3	6.8	3.2	1.5	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.6	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.5	0.6	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.6	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.6	0.6	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.6	0.8	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.5	0.7	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	2.5	0.8	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.8	2.5	0.7	0.4	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.1	48.1	22.8	11.5	3.5	1
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	52.7	20	9.1	1.7	1
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	47	24	11.5	4.5	0.9
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	52.2	21.1	9	1.8	1
U2 - Toxicity prior Beta(0.5, 0.5)	10.1	47.2	23.2	13.8	5	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	12.3	51.9	22	10.4	2.6	0.8
U2 - Toxicity prior Beta(1, 1)	9.6	47.1	23.5	14	5	0.8
U3 - Toxicity prior Beta(1, 1)	11.6	52	21.9	11	2.7	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.6	12.6	5.8	2.9	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	8	13.5	5.3	2.5	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	12.6	5.9	3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	7.8	13.3	5.4	2.6	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	6.5	12.6	6.1	3.4	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	6.9	13.5	5.8	2.8	0.8	
U2 - Toxicity prior Beta(1, 1)	6.3	12.7	6.2	3.4	1.2	
U3 - Toxicity prior Beta(1, 1)	6.7	13.7	5.7	2.9	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.4	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.5	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.4	1.3	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.5	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.3	0.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.1	0.6	0.2	
U2 - Toxicity prior Beta(1, 1)	0.3	3	1.4	0.8	0.3	
U3 - Toxicity prior Beta(1, 1)	0.3	3.1	1.1	0.6	0.2	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	44.6	16.2	3.1	0.9	0.3	34.9
U3 - Toxicity prior Beta(0.15, 0.85)	45.9	15.4	2.3	0.9	0.3	35.2
U2 - Toxicity prior Beta(0.3, 0.7)	43.5	18	3.7	1.5	0.4	32.9
U3 - Toxicity prior Beta(0.3, 0.7)	44.7	17.4	2.3	1.3	0.4	33.9
U2 - Toxicity prior Beta(0.5, 0.5)	44.3	18.9	5.3	2	0.7	28.8
U3 - Toxicity prior Beta(0.5, 0.5)	46.1	17.5	4.1	1.6	0.7	30
U2 - Toxicity prior Beta(1, 1)	44.1	18	6.3	2	0.8	28.8
U3 - Toxicity prior Beta(1, 1)	45.9	17.4	4.6	1.8	0.5	29.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.7	6.2	2.3	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	16	6	2.2	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	6.4	2.4	1.3	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	16	6.1	2.2	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	14.5	6.9	3.1	1.6	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	15.1	6.5	3	1.5	0.4	
U2 - Toxicity prior Beta(1, 1)	14.4	6.9	3.3	1.6	0.5	
U3 - Toxicity prior Beta(1, 1)	14.8	6.6	3.2	1.6	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	2.9	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	2.9	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	3	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	2.9	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1	2.8	0.8	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	1	2.9	0.8	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	1	2.9	0.8	0.5	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.7	15.3	22.3	16.8	26.5	4.4
U3 - Toxicity prior Beta(0.15, 0.85)	21.7	16.8	18.6	15.2	22.9	4.8
U2 - Toxicity prior Beta(0.3, 0.7)	13	13.9	22.1	16.9	30.1	4
U3 - Toxicity prior Beta(0.3, 0.7)	19.1	16.6	18.9	15.9	25.7	3.8
U2 - Toxicity prior Beta(0.5, 0.5)	12.6	12.3	22	16.8	33.8	2.5
U3 - Toxicity prior Beta(0.5, 0.5)	18.7	16.1	20.2	15.1	27.2	2.7
U2 - Toxicity prior Beta(1, 1)	12.6	12.7	21.7	17.1	33.3	2.6
U3 - Toxicity prior Beta(1, 1)	18.9	15.5	19.9	15.1	28.2	2.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7	6.1	5.9	5.5	5.1	
U3 - Toxicity prior Beta(0.15, 0.85)	7.6	6	5.8	5.4	4.8	
U2 - Toxicity prior Beta(0.3, 0.7)	6.7	5.8	6	5.5	5.5	
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	5.7	5.8	5.5	5.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.8	5.6	6.1	5.8	6.3	
U3 - Toxicity prior Beta(0.5, 0.5)	6.5	5.5	6.1	5.7	5.9	
U2 - Toxicity prior Beta(1, 1)	5.8	5.5	6.1	5.9	6.3	
U3 - Toxicity prior Beta(1, 1)	6.3	5.5	6.1	5.8	6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.6	1.9	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1.5	1.8	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.6	1.9	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.6	1.8	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.6	2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.6	1.9	1	
U2 - Toxicity prior Beta(1, 1)	0.5	3.1	1.6	2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.5	3.1	1.6	1.9	1	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	43.5	17.2	6.3	1.5	20.7
U3 - Toxicity prior Beta(0.15, 0.85)	12.1	46	15.5	4.7	0.9	20.8
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	43	18.8	7.4	2	19.7
U3 - Toxicity prior Beta(0.3, 0.7)	10.6	46.2	17.1	5.5	1.1	19.5
U2 - Toxicity prior Beta(0.5, 0.5)	9.9	43.7	19.9	8.8	2.4	15.3
U3 - Toxicity prior Beta(0.5, 0.5)	10.7	48.5	17.4	6.5	1.7	15.2
U2 - Toxicity prior Beta(1, 1)	8.8	44.8	19.8	9.1	2.5	15
U3 - Toxicity prior Beta(1, 1)	10.5	47.8	18.6	6.6	1.6	14.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.5	10.7	4.8	2.2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	11.1	4.6	2	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.3	10.7	5	2.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	11.1	4.8	2.2	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8	11.2	5.7	2.7	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8	11.9	5.5	2.4	0.7	
U2 - Toxicity prior Beta(1, 1)	7.7	11.4	5.8	2.7	0.9	
U3 - Toxicity prior Beta(1, 1)	7.9	11.9	5.5	2.5	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	4	1.1	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	4	1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	4	1.2	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	4.1	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.3	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.2	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	3.4	1.3	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.5	3.4	1.2	0.7	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66	23.5	5.3	0.5	0	4.7
U3 - Toxicity prior Beta(0.15, 0.85)	69.7	20.4	4	0.6	0	5.3
U2 - Toxicity prior Beta(0.3, 0.7)	63.2	25.9	5.5	0.8	0.1	4.5
U3 - Toxicity prior Beta(0.3, 0.7)	68.6	21.7	4.7	0.6	0.1	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	59.7	28	7.7	1	0.1	3.5
U3 - Toxicity prior Beta(0.5, 0.5)	64.7	25.9	5.2	0.7	0.1	3.4
U2 - Toxicity prior Beta(1, 1)	58.2	29	7.9	1.1	0.1	3.7
U3 - Toxicity prior Beta(1, 1)	64	26.4	5.3	0.7	0.1	3.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.5	7.9	3.4	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	17.1	7.6	3.2	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	16	8	3.6	1.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	16.8	7.7	3.2	1.4	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	14.6	8.5	4.2	1.7	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	15.3	8.3	3.9	1.6	0.4	
U2 - Toxicity prior Beta(1, 1)	14.4	8.4	4.4	1.8	0.4	
U3 - Toxicity prior Beta(1, 1)	15.2	8.3	4	1.6	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.5	1	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.4	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.5	1	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.3	1	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.4	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.4	1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	1	3.4	1.2	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	1.1	3.4	1.1	0.6	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.7	11.2	8.4	44.2	24.1	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	16.6	14.3	6.7	41	21.2	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	11.7	10.1	7.2	44.2	26.4	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	13.1	6.3	41.7	23	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	10.1	9.3	5.3	44.5	30.6	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	13.7	13.2	5.2	41.9	25.8	0.2
U2 - Toxicity prior Beta(1, 1)	9	9	6	45.5	30.3	0.2
U3 - Toxicity prior Beta(1, 1)	13.1	12.3	4.9	43.1	26.3	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	5.1	5.4	9.4	5.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4.9	5.1	5.3	9.6	5.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	4.8	5.2	9.8	5.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	4.9	5	9.8	5.5	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.5	4.9	10.2	6.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	4.7	4.8	10.3	5.8	
U2 - Toxicity prior Beta(1, 1)	4	4.5	4.9	10.3	6.3	
U3 - Toxicity prior Beta(1, 1)	4.3	4.6	4.8	10.3	5.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.5	2.3	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.1	1.5	2.3	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3	1.5	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3	1.5	2.2	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.3	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.5	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.5	2.3	1.3	
U3 - Toxicity prior Beta(1, 1)	0.3	2.9	1.4	2.2	1.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.2	44.8	17.1	6.7	1.7	17.5
U3 - Toxicity prior Beta(0.15, 0.85)	14.9	43.9	17	5.3	1.3	17.6
U2 - Toxicity prior Beta(0.3, 0.7)	12.2	42.9	18.7	7.8	2	16.4
U3 - Toxicity prior Beta(0.3, 0.7)	14.1	44.2	17.8	5.7	1.6	16.6
U2 - Toxicity prior Beta(0.5, 0.5)	11.9	45.1	19.7	8.4	2.5	12.4
U3 - Toxicity prior Beta(0.5, 0.5)	14.2	47.3	17.4	6.3	1.8	13
U2 - Toxicity prior Beta(1, 1)	11	46.2	19.5	8.2	2.9	12.2
U3 - Toxicity prior Beta(1, 1)	13.6	48.3	17.6	6.8	1.8	11.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.5	9.9	5	3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	10.2	4.9	2.7	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	9.2	9.8	5.3	3.1	1	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	10.1	5.1	2.8	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	10.3	5.9	3.5	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	10.7	5.7	3.2	1	
U2 - Toxicity prior Beta(1, 1)	7.6	10.5	6	3.5	1.3	
U3 - Toxicity prior Beta(1, 1)	8	10.8	5.8	3.2	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.2	1.3	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.2	1.2	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4.2	1.4	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4.2	1.3	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.8	1.5	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	1.4	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.6	1.5	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.6	3.7	1.4	1	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.6	17.6	13.7	34.4	14.9	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	26.8	18.9	11.6	30.2	11.6	0.9
U2 - Toxicity prior Beta(0.3, 0.7)	17.2	16.8	13.8	34.9	16.8	0.5
U3 - Toxicity prior Beta(0.3, 0.7)	24.8	19.9	10.3	31	13.4	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	15.8	15.9	12.9	35.2	19.7	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	23.9	18.9	10.9	30.9	14.7	0.7
U2 - Toxicity prior Beta(1, 1)	15.3	16	13.1	35.1	20	0.5
U3 - Toxicity prior Beta(1, 1)	23.8	19	10.2	31.1	15.3	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.2	6.1	5.6	7.7	4.3	
U3 - Toxicity prior Beta(0.15, 0.85)	6.8	6	5.4	7.7	4	
U2 - Toxicity prior Beta(0.3, 0.7)	5.8	5.8	5.4	8.1	4.7	
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	5.8	5.2	8	4.4	
U2 - Toxicity prior Beta(0.5, 0.5)	5.4	5.6	5.4	8.4	5.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5.9	5.5	5.2	8.4	4.9	
U2 - Toxicity prior Beta(1, 1)	5.3	5.5	5.4	8.5	5.3	
U3 - Toxicity prior Beta(1, 1)	5.8	5.5	5.2	8.4	5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.5	1.6	2.2	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	1.6	2.1	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.4	1.6	2.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.4	1.5	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.6	2.3	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.2	1.5	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.5	3.2	1.6	2.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.6	3.2	1.5	2.2	1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	47.7	30.2	13.8	6.1	0.6	1.6
U3 - Toxicity prior Beta(0.15, 0.85)	61.3	26.2	8	2.5	0.4	1.6
U2 - Toxicity prior Beta(0.3, 0.7)	49.1	30.9	12.1	5.9	0.7	1.3
U3 - Toxicity prior Beta(0.3, 0.7)	58.1	28.9	7.9	3.3	0.4	1.4
U2 - Toxicity prior Beta(0.5, 0.5)	46.6	29.3	15.7	6.4	1	1
U3 - Toxicity prior Beta(0.5, 0.5)	57.7	28.3	8.6	3.8	0.6	1
U2 - Toxicity prior Beta(1, 1)	48.1	28.9	14.5	6.6	1.1	0.8
U3 - Toxicity prior Beta(1, 1)	57.8	27.7	9	4	0.5	1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	8.2	5.7	4.2	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	11.1	7.9	5.4	4	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.8	8.1	5.9	4.4	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	7.8	5.5	4.2	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	8.1	6.2	4.6	2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.7	7.9	5.8	4.5	1.9	
U2 - Toxicity prior Beta(1, 1)	8.8	8.1	6.3	4.7	2	
U3 - Toxicity prior Beta(1, 1)	9.6	8	5.9	4.4	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.1	1.8	1.6	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	4	1.7	1.5	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4	1.8	1.7	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.9	1.8	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.8	1.9	1.7	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.8	1.8	1.7	0.5	
U2 - Toxicity prior Beta(1, 1)	1	3.8	1.8	1.8	0.5	
U3 - Toxicity prior Beta(1, 1)	1.2	3.8	1.8	1.7	0.5	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	41.3	25	13.3	13.7	6.7	0
U3 - Toxicity prior Beta(0.15, 0.85)	46.9	24.5	13.3	11.2	4.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	41.5	23.7	13.5	13.6	7.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	46.8	24.2	13	11.4	4.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	41	24.9	13.4	12.4	8.3	0
U3 - Toxicity prior Beta(0.5, 0.5)	45.9	25	13.8	10.7	4.6	0
U2 - Toxicity prior Beta(1, 1)	41.2	23.9	13.6	12.2	9.1	0
U3 - Toxicity prior Beta(1, 1)	45.4	25.1	13.6	10.9	5	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.5	4.9	6	8.3	6.4	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	5.1	6.2	8.3	5.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	4.8	6	8.3	6.5	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	5.1	6.2	8.3	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	4.7	5.9	8.4	6.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	5	6.3	8.4	6	
U2 - Toxicity prior Beta(1, 1)	4.2	4.8	5.9	8.3	6.8	
U3 - Toxicity prior Beta(1, 1)	4.3	5	6.2	8.4	6.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2	2.9	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	2.1	2.9	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	2	2.9	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	2.1	2.9	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.9	2.9	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	2.1	2.9	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	2	2.9	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	2.1	2.9	1.4	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.8	58.9	17.3	2.8	0.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	24.4	60	13.7	1.8	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	19.6	59.4	18	2.8	0.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	22.8	61.1	14	2	0	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	18.1	58	20.4	3.2	0.1	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	21.4	59.6	16.6	2.3	0	0.1
U2 - Toxicity prior Beta(1, 1)	18.5	57.8	20.5	3	0.1	0.1
U3 - Toxicity prior Beta(1, 1)	21.3	60.2	16.2	2.1	0	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	14.1	5.3	1.3	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.8	14.4	4.7	1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.8	14	5.5	1.4	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	14.5	4.7	1.1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.1	14	6	1.6	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	14.6	5.2	1.3	0.2	
U2 - Toxicity prior Beta(1, 1)	8.1	13.9	6.1	1.6	0.2	
U3 - Toxicity prior Beta(1, 1)	8.7	14.6	5.3	1.3	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.7	1.2	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.7	1	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.3	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.1	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.4	3.5	1.3	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	3.5	1.1	0.4	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.6	11.3	7.5	14.5	55.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	17.1	13.5	9.1	12.3	48	0
U2 - Toxicity prior Beta(0.3, 0.7)	11.4	10.7	7.5	14.4	56	0
U3 - Toxicity prior Beta(0.3, 0.7)	17	13.1	8	11.6	50.3	0
U2 - Toxicity prior Beta(0.5, 0.5)	10.4	9.2	7.5	12.6	60.3	0
U3 - Toxicity prior Beta(0.5, 0.5)	16.9	13	8.5	10.4	51.2	0
U2 - Toxicity prior Beta(1, 1)	10.5	8.5	8.3	12.9	59.8	0
U3 - Toxicity prior Beta(1, 1)	16.6	12.4	7.8	10.3	52.9	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4	5.1	6.8	10.6	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.1	5.2	6.9	10.2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	3.9	5	6.8	10.8	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4.1	5.1	6.8	10.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.8	4.8	6.8	11.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.9	5	6.9	10.8	
U2 - Toxicity prior Beta(1, 1)	3.3	3.7	4.8	6.8	11.4	
U3 - Toxicity prior Beta(1, 1)	3.4	3.9	4.9	6.9	10.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.5	2.5	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.4	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.5	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.4	1.6	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.5	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.5	59	20.7	3.1	0.1	2.6
U3 - Toxicity prior Beta(0.15, 0.85)	17.7	61.1	16	2.5	0.1	2.6
U2 - Toxicity prior Beta(0.3, 0.7)	12.8	59.7	22.6	2.7	0.1	2.1
U3 - Toxicity prior Beta(0.3, 0.7)	16.7	60.8	17.8	2.8	0.1	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	13.1	58	24.6	2.9	0.2	1.2
U3 - Toxicity prior Beta(0.5, 0.5)	15.8	59.1	20.7	3	0.1	1.3
U2 - Toxicity prior Beta(1, 1)	13	58.6	24	3.3	0.1	1
U3 - Toxicity prior Beta(1, 1)	15.5	59.2	21	3	0.2	1.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8	12.6	6.6	2.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.4	12.8	6.2	2.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	7.6	12.4	6.9	2.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	7.9	12.6	6.5	2.4	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	6.8	12.4	7.3	2.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	7.2	12.7	6.9	2.6	0.6	
U2 - Toxicity prior Beta(1, 1)	6.5	12.5	7.4	2.8	0.7	
U3 - Toxicity prior Beta(1, 1)	7.1	12.6	7	2.6	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.6	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4.6	1.6	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.6	1.8	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4.5	1.7	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	4.1	1.8	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	4.1	1.7	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.6	4.1	1.9	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	4.1	1.7	0.9	0.2	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.8	48.5	27.5	12.1	1.8	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	14	50.4	23.6	10.4	1.3	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	45.4	29.6	12.5	2.1	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	48.5	24.3	10.2	1.7	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	11.5	43.8	28.7	12.7	3	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	15.1	48.1	23.7	10.5	2.2	0.4
U2 - Toxicity prior Beta(1, 1)	10.7	43.2	29.7	12.7	3.4	0.3
U3 - Toxicity prior Beta(1, 1)	14.5	47.4	24.9	10.4	2.5	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.8	10	8.2	5.2	1.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.3	10.4	7.8	4.8	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	9.4	8.2	5.6	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	5.1	9.9	7.9	5.2	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	9.1	8.1	6.1	2.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.6	9.7	7.9	5.6	2.2	
U2 - Toxicity prior Beta(1, 1)	4.2	9.1	8.1	6.1	2.5	
U3 - Toxicity prior Beta(1, 1)	4.6	9.6	7.9	5.7	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.9	2.3	1.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.8	2.2	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.7	2.3	1.8	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	2.3	1.7	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.5	2.3	1.9	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	2.2	1.7	0.6	
U2 - Toxicity prior Beta(1, 1)	0.3	3.4	2.3	1.8	0.7	
U3 - Toxicity prior Beta(1, 1)	0.4	3.5	2.2	1.8	0.6	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	57.1	1.8	0	0	0	41.1
U3 - Toxicity prior Beta(0.15, 0.85)	57.9	1.5	0	0	0	40.6
U2 - Toxicity prior Beta(0.3, 0.7)	58.2	1.8	0	0	0	40
U3 - Toxicity prior Beta(0.3, 0.7)	58.3	1.4	0	0	0	40.3
U2 - Toxicity prior Beta(0.5, 0.5)	58.6	2	0	0	0	39.4
U3 - Toxicity prior Beta(0.5, 0.5)	59.2	1.8	0	0	0	39
U2 - Toxicity prior Beta(1, 1)	58.5	2	0	0	0	39.5
U3 - Toxicity prior Beta(1, 1)	58.9	1.6	0	0	0	39.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.2	2.5	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.4	2.5	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.3	2.6	0.3	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.4	2.5	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.1	2.8	0.4	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	2.7	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	20	2.9	0.4	0	0	
U3 - Toxicity prior Beta(1, 1)	20.2	2.8	0.3	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	1.1	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.1	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.1	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.3	1.1	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.3	1.1	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.4	62.5	14.1	3.7	0	1.3
U3 - Toxicity prior Beta(0.15, 0.85)	23	60.5	11.4	3.5	0	1.6
U2 - Toxicity prior Beta(0.3, 0.7)	17.3	63.1	14.8	4	0	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	22.3	60.1	12.3	4.3	0	1
U2 - Toxicity prior Beta(0.5, 0.5)	17.5	59.8	16.3	5.4	0	1
U3 - Toxicity prior Beta(0.5, 0.5)	20.4	60.3	13.8	4.4	0	1.1
U2 - Toxicity prior Beta(1, 1)	16.3	60.7	16.9	5	0	1.1
U3 - Toxicity prior Beta(1, 1)	20.9	58.3	15	4.7	0	1.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.5	13.3	5.7	2.2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	13.1	5.4	2.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.2	13	6	2.4	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	13	5.7	2.3	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	7.5	12.9	6.4	2.8	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8	12.9	6	2.6	0.3	
U2 - Toxicity prior Beta(1, 1)	7.4	12.9	6.4	2.9	0.4	
U3 - Toxicity prior Beta(1, 1)	8	12.8	6.1	2.7	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.8	1.7	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.7	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4.7	1.7	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.5	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	4.4	1.8	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.3	1.7	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	4.4	1.8	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	4.3	1.7	0.7	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21	4.2	0	0	0	74.8
U3 - Toxicity prior Beta(0.15, 0.85)	20.9	3.9	0.1	0	0	75.1
U2 - Toxicity prior Beta(0.3, 0.7)	21	5.4	0	0	0	73.6
U3 - Toxicity prior Beta(0.3, 0.7)	21	5.5	0.1	0	0	73.4
U2 - Toxicity prior Beta(0.5, 0.5)	21.9	6.5	0	0	0	71.6
U3 - Toxicity prior Beta(0.5, 0.5)	20.9	7.4	0.3	0	0	71.4
U2 - Toxicity prior Beta(1, 1)	22.1	6.1	0.1	0	0	71.7
U3 - Toxicity prior Beta(1, 1)	20.9	7.7	0	0	0	71.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.9	4.8	0.7	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.9	4.7	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.9	5	0.7	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	16.1	4.8	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.8	5.2	1.2	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	5	1.2	0.1	0	
U2 - Toxicity prior Beta(1, 1)	15.9	5.2	1.3	0.1	0	
U3 - Toxicity prior Beta(1, 1)	16	5	1.3	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.8	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.7	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.9	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.7	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.7	0.3	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.6	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	0.8	2.7	0.3	0	0	
U3 - Toxicity prior Beta(1, 1)	0.8	2.6	0.3	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	2.3	1.4	3.1	56.3	36.9	0
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	3	4.8	58.6	29.7	0
U2 - Toxicity prior Beta(0.3, 0.7)	2.4	1.2	3.1	56.6	36.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	3	4.3	58.6	30.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	2.3	1.3	3.4	55.2	37.8	0
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	2.5	4.5	58.7	30.8	0
U2 - Toxicity prior Beta(1, 1)	2.2	1.3	3	54.9	38.6	0
U3 - Toxicity prior Beta(1, 1)	3.4	2.8	4.6	57.6	31.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.1	4	12.5	7.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	4.1	13.1	6.5	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.1	3.9	12.6	7.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	4.1	13.1	6.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	4	12.5	7.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	4	13.1	6.7	
U2 - Toxicity prior Beta(1, 1)	3.1	3.1	3.9	12.6	7.3	
U3 - Toxicity prior Beta(1, 1)	3.1	3.1	4	13	6.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.5	1.1	2.5	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1	2.5	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.4	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.2	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.4	1.2	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	28.1	28.6	37	6.1	0.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	32.8	30.6	32	4.5	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	28	27.3	38.2	6.1	0.2	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	33.5	26.5	34	5.7	0.1	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	26.4	25.9	40.4	6.9	0.3	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	32.4	25.6	35.7	6	0.2	0.1
U2 - Toxicity prior Beta(1, 1)	26.4	26.3	40.4	6.6	0.2	0.1
U3 - Toxicity prior Beta(1, 1)	31.5	27	35.4	5.8	0.2	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	9	8.8	2.8	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	9.2	8.3	2.5	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	8.7	9.1	3.1	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.3	8.9	8.6	2.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7.8	8.6	9.6	3.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	8.7	9.2	3.1	0.5	
U2 - Toxicity prior Beta(1, 1)	7.7	8.6	9.7	3.4	0.6	
U3 - Toxicity prior Beta(1, 1)	8.3	8.8	9.2	3.2	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.7	1.9	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.6	1.8	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.6	2	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.8	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	2	1	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.4	1.9	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.4	2	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	3.4	1.8	1	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.8	32.2	34.1	7.4	0.1	3.4
U3 - Toxicity prior Beta(0.15, 0.85)	28.6	31.3	30.2	6.5	0.1	3.3
U2 - Toxicity prior Beta(0.3, 0.7)	21.5	30.2	36.7	8.7	0.1	2.8
U3 - Toxicity prior Beta(0.3, 0.7)	28.6	29.7	31.9	7	0.1	2.7
U2 - Toxicity prior Beta(0.5, 0.5)	19.7	29.7	39.8	8.6	0.2	2
U3 - Toxicity prior Beta(0.5, 0.5)	27.3	28.4	34.3	7.9	0.1	2
U2 - Toxicity prior Beta(1, 1)	18.9	29.8	39.9	9.2	0.3	1.9
U3 - Toxicity prior Beta(1, 1)	26.7	28.4	35.3	7.7	0.1	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	8.9	8.4	3.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.4	8.8	8.1	2.9	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.5	8.7	8.6	3.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9	8.6	8.4	3.2	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	7.4	8.7	9.3	3.8	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	7.9	8.7	9.1	3.5	0.5	
U2 - Toxicity prior Beta(1, 1)	7.2	8.7	9.5	3.7	0.6	
U3 - Toxicity prior Beta(1, 1)	7.8	8.7	9.2	3.5	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.7	1.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.9	1.6	1.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.9	1.7	1.2	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.9	1.6	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	1.8	1.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.7	1.2	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.6	1.8	1.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.6	1.8	1.2	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.6	26.6	46.7	11	0.5	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	20.7	26.4	42.6	9.1	0.4	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	14.5	23.9	48.2	12.4	0.5	0.5
U3 - Toxicity prior Beta(0.3, 0.7)	19.5	25.8	43.7	10	0.4	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	12.4	22.6	49.9	13.6	1.1	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	17.7	23.7	45.8	11.4	0.9	0.5
U2 - Toxicity prior Beta(1, 1)	12.3	22.9	49.3	13.9	1.3	0.3
U3 - Toxicity prior Beta(1, 1)	16.9	25.4	44.1	12.2	1.1	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.6	8.2	10.2	4	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	7.2	8.1	10.1	3.8	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	6.4	7.9	10.3	4.4	1	
U3 - Toxicity prior Beta(0.3, 0.7)	6.9	7.8	10.2	4.1	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	5.5	7.6	10.8	4.8	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	6	7.7	10.8	4.5	1	
U2 - Toxicity prior Beta(1, 1)	5.4	7.6	10.9	4.9	1.2	
U3 - Toxicity prior Beta(1, 1)	5.8	7.7	10.9	4.7	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.7	2.1	1.4	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.7	2	1.4	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	2.1	1.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.6	2	1.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	2.1	1.5	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	2	1.4	0.2	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	2.1	1.5	0.3	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	2	1.5	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.7	52	9.4	0.5	0	19.4
U3 - Toxicity prior Beta(0.15, 0.85)	20.6	51.6	7.8	0.3	0	19.7
U2 - Toxicity prior Beta(0.3, 0.7)	17.8	53.7	10.7	0.5	0	17.3
U3 - Toxicity prior Beta(0.3, 0.7)	18.2	53.8	9.3	0.5	0	18.2
U2 - Toxicity prior Beta(0.5, 0.5)	15.6	58.1	10.5	1	0	14.8
U3 - Toxicity prior Beta(0.5, 0.5)	17	58.3	9.3	0.7	0	14.7
U2 - Toxicity prior Beta(1, 1)	14	60.5	10.2	1.1	0	14.2
U3 - Toxicity prior Beta(1, 1)	17.4	57.5	9.5	0.7	0	14.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.7	12.1	4.1	1.2	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11	12	3.9	1.1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.4	12.3	4.3	1.3	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	12.2	4.2	1.2	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	9.2	12.9	4.9	1.5	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	9.5	13.1	4.6	1.3	0.2	
U2 - Toxicity prior Beta(1, 1)	9.1	13	5	1.5	0.3	
U3 - Toxicity prior Beta(1, 1)	9.5	13.1	4.7	1.4	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.6	1.2	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.5	1.2	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.6	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.5	1.2	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.2	1.4	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.2	1.3	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.9	4.2	1.4	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	4.1	1.3	0.4	0.1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66.1	27.1	4.2	0.2	0	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	69.6	25.2	2.8	0.1	0	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	64.4	28.3	5.1	0	0	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	67.2	26.1	4.3	0	0	2.4
U2 - Toxicity prior Beta(0.5, 0.5)	60.3	31.6	5.9	0.2	0	2
U3 - Toxicity prior Beta(0.5, 0.5)	65	28.9	3.9	0.1	0	2.1
U2 - Toxicity prior Beta(1, 1)	60.6	31	6.2	0.2	0	2
U3 - Toxicity prior Beta(1, 1)	65.8	27.8	4.3	0	0	2.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.7	8.5	2.7	0.6	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.7	8.2	2.2	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.5	8.6	2.8	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	18.4	8.3	2.2	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	16.4	9.3	3.2	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	9	2.6	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	16.3	9.3	3.2	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	17.2	9	2.7	0.6	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3	0.9	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.9	0.7	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3	0.9	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.9	0.7	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3	1	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	0.8	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.8	3	1	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.8	2.8	0.8	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.2	11.9	11.2	45.3	14.4	0
U3 - Toxicity prior Beta(0.15, 0.85)	27.9	14	9.8	38.3	10	0
U2 - Toxicity prior Beta(0.3, 0.7)	16.9	10.9	11.6	45.4	15.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	25.9	13.7	9.8	39.4	11.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	15.7	10.9	10	46.5	16.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	23.6	13.7	9.6	39.4	13.7	0
U2 - Toxicity prior Beta(1, 1)	16.4	11.7	10.8	44.2	16.9	0
U3 - Toxicity prior Beta(1, 1)	24.1	13.3	9.5	39.7	13.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	4.8	6	10.3	4.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	4.9	5.9	10.3	4.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	4.6	5.8	10.5	4.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	4.8	5.9	10.5	4.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.5	5.5	10.8	5.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	4.6	5.7	10.7	4.8	
U2 - Toxicity prior Beta(1, 1)	3.9	4.5	5.6	10.7	5.3	
U3 - Toxicity prior Beta(1, 1)	4.1	4.6	5.7	10.7	4.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.7	2.4	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.7	2.4	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.7	2.4	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.7	2.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.6	2.3	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.6	2.3	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.6	2.3	1.3	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.6	2.3	1.2	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	35.7	26.7	26.2	6.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	7.7	42.5	25.4	20.2	4.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	5.5	34.5	27	26.5	6.5	0
U3 - Toxicity prior Beta(0.3, 0.7)	7.9	41.8	25.5	20.3	4.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	5.2	35.3	26.1	26	7.4	0
U3 - Toxicity prior Beta(0.5, 0.5)	7.4	42.2	25.1	20.3	5	0
U2 - Toxicity prior Beta(1, 1)	5.8	35	26	26.3	6.9	0
U3 - Toxicity prior Beta(1, 1)	7.8	42.4	23.9	21.2	4.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	11.1	7.2	5.8	2.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	12	7.1	5.3	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	11	7.2	5.9	2.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	12	7.1	5.4	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	10.9	7.2	5.8	2.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	11.9	7.1	5.4	1.7	
U2 - Toxicity prior Beta(1, 1)	3.7	10.9	7.2	5.9	2.2	
U3 - Toxicity prior Beta(1, 1)	3.9	11.9	7.1	5.4	1.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.1	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	1.1	0.5	
U2 - Toxicity prior Beta(1, 1)	0	2.6	1.5	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	1.1	0.5	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.6	15.1	33.3	26.2	11.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	20	16.3	33.3	23.4	7	0
U2 - Toxicity prior Beta(0.3, 0.7)	13.6	14.8	33.8	25.9	11.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	20.2	16.1	33.8	22.5	7.4	0
U2 - Toxicity prior Beta(0.5, 0.5)	13.7	14.2	33.8	25.2	13.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	19.3	16.2	34.2	23.2	7.1	0
U2 - Toxicity prior Beta(1, 1)	12.7	14.8	33.2	26.6	12.7	0
U3 - Toxicity prior Beta(1, 1)	19.5	16.1	32.1	25	7.3	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4.3	8.5	8.5	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.5	8.9	8.4	4.6	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	4.1	8.3	8.5	5.6	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4.3	8.7	8.6	4.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	4.1	8.1	8.8	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	4.2	8.6	8.7	5	
U2 - Toxicity prior Beta(1, 1)	3.4	4	8	8.8	5.8	
U3 - Toxicity prior Beta(1, 1)	3.5	4.2	8.5	8.8	5.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.9	2.7	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.8	2.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.7	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.8	2.8	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.9	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.8	2.8	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.8	2.8	1	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.3	5.1	10.4	45.1	31.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	12.6	7.5	12.7	43	24.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	8	4.8	11	44.8	31.4	0
U3 - Toxicity prior Beta(0.3, 0.7)	12.5	6.8	12.8	42.3	25.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	3.9	10.7	45.1	34.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	11.4	6.3	11.7	42.6	28	0
U2 - Toxicity prior Beta(1, 1)	6.3	4	9.3	45.8	34.6	0
U3 - Toxicity prior Beta(1, 1)	10.9	6.1	10.9	43.3	28.8	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.6	5.9	10.3	6.8	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	6	10.5	6.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.6	5.7	10.5	7	
U3 - Toxicity prior Beta(0.3, 0.7)	3.4	3.7	5.8	10.6	6.4	
U2 - Toxicity prior Beta(0.5, 0.5)	3.2	3.5	5.4	10.7	7.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.6	5.5	10.9	6.6	
U2 - Toxicity prior Beta(1, 1)	3.2	3.5	5.3	10.6	7.3	
U3 - Toxicity prior Beta(1, 1)	3.3	3.6	5.5	10.9	6.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.5	2.8	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.4	2.7	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.5	2.6	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.6	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.6	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.6	1.3	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.6	1.2	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.4	12.8	44.8	22.3	5.5	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	17.9	15.2	43	20.4	3.3	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	12.4	12.3	45.5	23.3	6.3	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	18.4	14.5	42.7	21.1	3.1	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	12.5	10.6	44.5	24.9	7.3	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	18.2	15.9	42	20.1	3.5	0.3
U2 - Toxicity prior Beta(1, 1)	11.1	11.2	45.3	24.7	7.4	0.3
U3 - Toxicity prior Beta(1, 1)	17	14.8	42.5	21.1	4.3	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.1	6.3	9.6	6.2	2.8	
U3 - Toxicity prior Beta(0.15, 0.85)	5.8	6.3	9.5	6	2.4	
U2 - Toxicity prior Beta(0.3, 0.7)	5	6	9.4	6.5	3.1	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	6	9.4	6.3	2.7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.6	5.7	9.2	7	3.5	
U3 - Toxicity prior Beta(0.5, 0.5)	5	5.8	9.3	6.8	3	
U2 - Toxicity prior Beta(1, 1)	4.5	5.6	9.3	7	3.5	
U3 - Toxicity prior Beta(1, 1)	5	5.8	9.3	6.8	3.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.4	2.1	2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.5	2	1.9	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.3	2	2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	2	1.9	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	2	2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	2	1.9	0.6	
U2 - Toxicity prior Beta(1, 1)	0.4	3.1	2	2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.5	3.2	2	1.9	0.6	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.2	40.3	31.5	14.3	3.7	0
U3 - Toxicity prior Beta(0.15, 0.85)	13.8	46.6	27.7	9.4	2.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	39.7	31.4	15.3	4.2	0
U3 - Toxicity prior Beta(0.3, 0.7)	14.1	45.2	27.6	10	3.1	0
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	39	29.4	17.5	4.6	0
U3 - Toxicity prior Beta(0.5, 0.5)	13.6	43.7	27	12.2	3.4	0.1
U2 - Toxicity prior Beta(1, 1)	9.4	38.1	30.3	17.3	4.8	0.1
U3 - Toxicity prior Beta(1, 1)	13.9	43	27.6	11.9	3.5	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	11.2	8.5	4.4	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	12.2	7.9	4	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	10.9	8.3	4.7	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	11.8	8	4.2	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	10.8	8.2	5	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	11.6	7.9	4.5	1.5	
U2 - Toxicity prior Beta(1, 1)	4.2	10.6	8.3	5	1.9	
U3 - Toxicity prior Beta(1, 1)	4.4	11.6	8	4.4	1.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.9	1.2	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.8	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.9	1.2	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.8	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.8	1.3	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.8	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.9	1.2	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.8	1.1	0.3	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.8	48.7	15.4	7.8	1.6	5.7
U3 - Toxicity prior Beta(0.15, 0.85)	25	47.6	13	7.2	1.2	6
U2 - Toxicity prior Beta(0.3, 0.7)	17.6	47.9	16.2	10.6	2.2	5.5
U3 - Toxicity prior Beta(0.3, 0.7)	23	48	13.3	9.1	1.2	5.4
U2 - Toxicity prior Beta(0.5, 0.5)	15.5	47.2	19.9	10.9	3.1	3.4
U3 - Toxicity prior Beta(0.5, 0.5)	20.8	47.7	16.6	9.1	1.7	4.1
U2 - Toxicity prior Beta(1, 1)	14.4	48.2	20	10.7	2.9	3.8
U3 - Toxicity prior Beta(1, 1)	21.2	48.1	16.8	8.4	1.8	3.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	10.5	5.5	3.5	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	10.3	5.4	3.4	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	10.3	5.7	3.8	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	10.3	5.5	3.6	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	7.4	10.1	6.3	4.2	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	7.9	10.1	6.1	4	1.5	
U2 - Toxicity prior Beta(1, 1)	7.3	10.1	6.3	4.2	1.7	
U3 - Toxicity prior Beta(1, 1)	7.8	10.1	6.2	4.1	1.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.6	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.6	1.5	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.6	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.5	1.6	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	4.2	1.7	1.3	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	4.1	1.6	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	0.7	4.1	1.7	1.3	0.4	
U3 - Toxicity prior Beta(1, 1)	0.8	4.1	1.6	1.3	0.4	

Phase I/II Operating Characteristics

Alpha model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.9	57.1	16.3	3.8	0.5	3.4
U3 - Toxicity prior Beta(0.15, 0.85)	24.7	55.3	13	3.1	0.5	3.4
U2 - Toxicity prior Beta(0.3, 0.7)	18.3	56.8	16.6	4.7	1	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	22.5	55.7	13.5	4.5	0.7	3.1
U2 - Toxicity prior Beta(0.5, 0.5)	17.9	54.6	18.9	5.5	1.1	2
U3 - Toxicity prior Beta(0.5, 0.5)	23	53.5	15.5	4.8	0.7	2.5
U2 - Toxicity prior Beta(1, 1)	16.8	55.4	19.1	5.3	1.1	2.3
U3 - Toxicity prior Beta(1, 1)	23.5	53.3	15.6	4.7	0.7	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	11.7	5.7	3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	11.4	5.4	3	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	11.6	6	3.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	8.7	11.3	5.7	3.2	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	11.4	6.4	3.6	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	7.9	11.2	6.1	3.5	1.2	
U2 - Toxicity prior Beta(1, 1)	7.2	11.5	6.3	3.6	1.3	
U3 - Toxicity prior Beta(1, 1)	7.9	11.2	6	3.5	1.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	5	1.7	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.9	1.6	1.1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.8	1.7	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.8	1.7	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	4.5	1.8	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.4	1.7	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	0.8	4.5	1.7	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	1	4.4	1.7	1.2	0.3	